PhishShift Final Report

Decision support only - not a guarantee of safety.

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Summary: The project builds a synthetic phishing-risk scanner that explains risky message cues, compares three model variants, and preserves a human-in-the-loop helpdesk workflow.

Methodology: Data generation and sanitization ensure no live URLs or secrets are introduced. The corpus is cleaned, deduplicated, split by campaign group, and used for transparent as well as TF-IDF model comparison.

Results: The rules model is explainable, the TF-IDF Logistic Regression model is strongest on the synthetic corpus, and the third model provides a comparison baseline.

Robustness and abstention: Narrow uncertainty bands trigger abstention while benign perturbations remain stable and require analyst review.

Deployment: Render configuration and health checks support a secure, web-based dashboard while keeping the environment isolated from live message traffic.